List Comprehension

```
A compact way to create a list from a sequence

- list(string)
# i for i in string
- [i for i in range(11)]
- [i*i for i in range(11)]
- [(i, i*i) for i in range(11)]
```

List Comprehension with If Expression

```
[i for i in range(21) if i % 2 == 0]
```

Flattening 3 Dimensional Array

```
[number for row in list for number in row]
```

Lambda function

```
A small anonymous function, can only have one expression, can take many
arguments, write function inside another function
```

| Creating Lambda Function

```
x = lambda param1, param2: param1 + param2
print(x(2, 3)) # 5
```

| Lambda Inside Function

```
def function(x):
    return lambda n: x**n
function(x)(n) # need two arguments
    # called as currying
```